SBMS COMPREHENSIVE INFORMATION REPORT FOR CHAPTERS FOUR, FIVE AND SIX

Document type: evidence dossier and drafting resource **System:** Student Beneficiary Management System (SBMS) **Revision described:** `fix/sbms-completion-work` @ `d6b5032` **Prepared:** 2026-11-02 **Institution context:** capstone project report; Design Science Research through an Agile cycle

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Document control

Item	Value
This report	`SBMS_COMPREHENSIVE_CHAPTERS_4_5_6_INFORMATION_REPORT.md` and `.pdf`
Ten supporting data files	`SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md`, `SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md`, `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md`, `SBMS_API_ENDPOINT_CATALOGUE.md`, `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md`, `SBMS_TEST_EVIDENCE_CATALOGUE.md`, `SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md`, `SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md`, `SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md`, `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`
Earlier deliverables re-used, not duplicated	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md`, `CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md`, `CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md`, `OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md`, `DOCUMENTATION_GAP_REGISTER.md`, `EVIDENCE_AND_VERIFICATION_REPORT.md`, `FIGURE_PLACEHOLDER_CHECKLIST.md`
Evidence position	Implementation facts read from code at the named revision; test counts taken from dated run records and **not re-executed**; ML metrics synthetic only; the unsupported claims are stated in Part III §24 and itemised in `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §4
What this document is not	It is not a replacement for the three chapters. It is the evidence base and drafting aid for them

How to use this report

1. **Part I** supplies everything Chapter Four needs: scope, actors, requirement counts, the analysis and design facts each diagram must encode, the interface inventory, the schema, and the deployment architecture.
2. **Part II** supplies everything Chapter Five needs: the implementation environment, the feature-by-feature implementation map, the test inventory with what each suite proves, the defect log, and the evaluation against objectives and requirements.
3. **Part III** supplies everything Chapter Six needs: an objective-by-objective conclusion with an honest status, the stated contribution, recommendations, and future work with owners.
4. **Part IV** is the assurance layer: a completeness matrix (requested area → status → where it is), the gap summary, reproduction commands and the document map.
5. Every factual statement names its source. Where a statement rests on code reading only, it says so. Where evidence is absent, the text says *not verified* rather than implying success.

Executive summary of the evidence position

Dimension	Position at this revision
Code	Complete three-tier system with a deployed demonstration instance; 45 HTTP endpoints, 13 tables, 14 forward migrations
Automated tests	151 Jest tests in 24 suites (2026-11-01), 43 native in-process tests, 3 live-HTTP smoke tests, 40 Python tests, 43 browser checks across three harnesses
Build	Frontend production build exits 0 (2 006 modules); API image built by CI and running on Azure
Machine learning	One deployed XGBoost artifact over six features; synthetic holdout F1 0.9063 at cut-off 0.50; **no real-world metric exists**
Deployment	Live on Azure (UAE North) as a demonstration instance, verified on 2026-10-31 and 2026-11-01

Dimension	Position at this revision
User evidence	**None.** No user acceptance study, no usability test, no fieldwork
Scale evidence	**None.** No load test; the only timing is 22.67 ms for 50 candidate pairs
Physical QR evidence	**None.** No phone-camera scan of a printed code was recorded

Three sentences summarise the honest position. The system is built, tested at API and browser level, and deployed. Its duplicate-detection model is evaluated only against synthetic data, so no accuracy claim about real beneficiaries is possible. Its usability, scale behaviour and physical QR identification are unverified and must be reported as such.

PART I — EVIDENCE FOR CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

1. System identity and problem framing

SBMS exists to replace a manual bursary register with a system that (a) registers a beneficiary once, (b) detects a probable repeat application before money is committed, (c) routes every duplicate signal to a human, and (d) leaves an audit trail from registration to receipt. The design's central control is therefore not the classifier: it is the **gate**. A model score may create a case, but only an administrator's decision can release an award for payment, and only a verified beneficiary may be approved at all.

Element	Fact at this revision
Presentation	React 18 SPA, Vite, `react-router-dom`; 14 screens across 16 routes
Application	Node 22 + Express 4; 13 routers, 13 controllers, 10 services, 5 models
Persistence	PostgreSQL 17; 13 tables; 14 forward migrations plus one rollback script
Inference	Python 3.11 + scikit-learn + XGBoost, one artifact, spawned as a stdio subprocess
Delivery	One container image (API plus ML) on Azure Container Apps; static frontend on App Service F1
Identity	JWT, 8-hour expiry, role re-read from the database on every request

The two material departures from the approved proposal must be stated in Chapter Four, not discovered in Chapter Six. First, the deployed classifier is **XGBoost over six features**, not the specified Random Forest over four; the Random Forest survives only as an unevaluated research candidate. Second, the system **exceeds** the proposal on award reconciliation and certificates while **falling short** of the proposal's implied evaluation design, because registration still compares a new record against the entire registry with no candidate blocking.

2. Requirements: what the chapter can state as fact

Group	Count	Verified	Partial	Not implemented
Functional requirements delivered (FR-01 to FR-30)	30	24	6	0
Proposal requirements carried as deviations (FR-31 to FR-33)	3	0	0	3
Functional gaps stated as gaps (FR-34 to FR-37)	4	0	0	4
Non-functional requirements (NFR-01 to NFR-21)	21	14	7	0

The register, with the statement, actor, priority, implementation location, test and status of each requirement, is `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §6-§7. Chapter Four should reproduce the register as its Table 4.1 and Table 4.2 rather than restating requirements in prose, because prose drifts and the register does not.

The six gaps that must appear as gaps: no candidate blocking (FR-34), no QR re-issue (FR-35), no token expiry (FR-36), no payment execution (FR-37), no NEMIS integration (FR-32), no offline or native mobile operation (FR-33).

3. Actors and their interests

Actor	Interest	Interface surface	Evidence of role enforcement
Field Officer	Register accurately, see whether a case was raised, record evidence, submit awards	`/students`, `/students/register`, `/eligibility`, `/disbursements`, `/field-officer`	403 on adjudication and approval (`workflow.test.js` #11)
Administrator	Decide cases, approve awards, control payments, issue certificates, read the audit trail	`/dashboard`, `/duplicate-review`, `/disbursements`, `/reports`, `/settings`, `/audit`, `/users`	403 on registration and award submission (`workflow.test.js` #11)
Public verifier	Confirm a printed code is genuine without learning anything about the holder	`/verify`, `/verify-certificate`	Boolean and authenticity facts only (15 tests)

Actor	Interest	Interface surface	Evidence of role enforcement
Operator	Apply migrations, manage secrets, start and stop the deployment, restore a backup	CLI, `deploy/azure/**`	Start-up guards; readiness contract
Programme owner	Set the cut-off policy, adjudication protocol and acknowledgement deadline	Not in the interface	Open decisions D1-D5

The role split is by *duty*, not seniority: the officer who registers a beneficiary cannot approve money for that beneficiary, and the administrator who approves cannot register. That separation is the design's answer to the collusion and error risk described in Chapter Two.

4. Analysis artefacts: what each diagram must encode

Chapter Four specifies eleven analysis artefacts. Each must be drawn from the implemented system, so each row gives the facts the drawing has to carry and the code that establishes them.

Figure	What it must show	Facts that must appear	Source
Context diagram (4.3.1)	SBMS and its external parties	Field officer, administrator, public verifier, PostgreSQL, the ML predictor as an internal component, no NEMIS link	`backend/src/index.js` mount table
Use case diagram (4.3.2)	Actors to use cases	The 14 screens' capabilities; public verification as unauthenticated	`frontend/src/App.jsx`; API catalogue §3
Activity: registration and screening (4.3.3)	Decision points	Validation, advisory lock, whole-registry scoring, qualification, `top-1` aggregation, 503 on predictor failure, student still created	`students.controller.js:31-161`
Activity: review to payment gate (4.3.4)	Two gates	Approval requires verified and no open or confirmed case; payment requires approved, verified, no open case and amount ceiling	`bursaryAwards.controller.js:151-209`; `disbursements.controller.js:65-136`
Class diagram (4.3.5)	Entities and services	Student, DuplicateCase, ReviewEvent, Award, VerificationEvent, Disbursement, ReconciliationEvent, Certificate, User	Data dictionary §4
Sequence: registration (4.3.6)	Message order across tiers	SPA → API → lock → SELECT FOR UPDATE → predictor → aggregation → insert → audit → response	Architecture map §5
Sequence: review decision (4.3.7)	Decision to terminal state	Masked queue, audited reveal, decision plus event plus audit in one transaction	`duplicateFlags.controller.js`
Sequence: award to certificate (4.3.8)	Four gates and a snapshot	Verification evidence, approval, payment ceiling, certificate snapshot with unique award	API catalogue §3.5-§3.6
Sequence: public verification (4.3.9)	Token handling	Fragment token, `replaceState`, single-key POST, existence query, boolean answer	`docs/qr-verification-2026-10-30.md` §1
ER diagram (4.3.10)	13 tables and relationships	Cascades for review data; RESTRICT for money, certification and verification records	Data dictionary §5
ML pipeline diagram (4.3.11)	Six stages	Generation, feature construction, training, validation, artifact, serving	ML documentation §2-§3

`FIGURE_PLACEHOLDER_CHECKLIST.md` already separates diagrams that can be drawn from specifications from those that need screenshots. No screenshot exists yet (G-04).

5. Design: architecture and module responsibilities

Layer	Responsibility	Notable design choice	Where the evidence sits
Presentation	Rendering, role routing, plain-language messaging	Officers never see model internals; administrators see score, version and cut-offs	UI inventory §3-§5

Layer	Responsibility	Notable design choice	Where the evidence sits
Application	Auth, validation, workflow, audit	The role is re-read from the database, so a stale token cannot carry an old privilege	Security catalogue §4
Domain services	Screening, aggregation, rules, documents, readiness	One shared module per rule set, so policy cannot drift between screens	Architecture map §3.2
Persistence	Constraints and referential actions	The database refuses an unverified approval and a half-recorded clearance	Data dictionary §7
Inference	Feature construction and scoring	Training and serving share one feature module	ML documentation §3

Three design decisions deserve explicit treatment because they are what the chapter's "design justification" section should argue:

1. **A human gate enforced by the schema.** `bursary_awards_approval_verification_check` makes it impossible to store an approved award for an unverified beneficiary, regardless of which code path attempted it. The control is not in the controller alone.
2. **Snapshot rather than reference for certificates.** A certificate copies the facts it prints, so a later award edit cannot silently alter what a printed document says; verification compares snapshot with live award and answers `not_current` on a mismatch.
3. **Aggregation of cases, not of scores.** `top-1` caps cases per registration at one while still counting suppressed candidates in the audit trail. It manages workload without pretending to improve the model.

6. Design: workflow and state

Entity	States	Transition enforcer	Refusal message
Duplicate case	`pending` → `confirmed_duplicate` / `not_duplicate` / `needs_verification` (terminal)	Controller plus event history	`This duplicate flag has already been decided`
Award	`submitted` → `approved` / `rejected`	Schema CHECK plus controller	`Verify the beneficiary before approving this award.` / `Resolve duplicate review cases before approving this award.`
Verification	`not_checked` → `verified` / `not_found` / `needs_investigation`	Schema CHECK on evidence	`Provide an outcome, notes, and evidence for verified cases.`
Payment	`issued` → `cleared` / `failed` / `cancelled`	`canTransitionPayment` plus CHECK	`This payment status cannot be changed.`
Receipt	`null` → `awaiting_acknowledgement` → `acknowledged` / `discrepancy`	`receiptStatus` in cents	`Enter the received amount and evidence reference.`
Certificate	`active` → `revoked`	CHECK plus UNIQUE award	`This certificate has already been revoked.`

A confirmed duplicate is deliberately **terminal**: it leaves the beneficiary unpayable and cannot be reversed through the API. That is correct policy, and it also means a mistaken confirmation can be corrected only in the database. The chapter should say so rather than describe the design as forgiving.

7. Design: access control, security and validation

Control	Implementation	What must be said about it
Authentication	bcrypt cost 12; JWT 8 h; secret >= 32 bytes validated at start-up	Implemented and tested; no multi-factor and no token revocation list
Authorisation	`requireRole` per route or router; role re-read from the row	Tested; matrix in the security catalogue §3
Account state	`is_active` checked on every request	Deactivation is the only revocation mechanism

Control	Implementation	What must be said about it
Origin control	Allowlist checked before routing plus `cors`	A browser policy, not authentication
Headers	`nosniff`, `DENY`, `no-referrer`, restricted permissions	No CSP and no HSTS from the API
Body bound	100 kB JSON limit	Tested
Error hygiene	5xx always `Internal server error`; allowlisted diagnostics only	Tested
Rate limits	10 sign-ins, 10 student verifications, 20 certificate verifications per IP per 15 min	In-process only; not shared across replicas
Validation	Zod per endpoint with cross-field rules	Tested through the API
Identifier masking	Last three characters until an audited reveal	Tested
Privacy of public endpoints	Boolean and authenticity facts only	15 tests

The residual risks in `SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md` §11 belong in the chapter's security-evaluation subsection. They are: a JWT in `localStorage`, no revocation list, no per-account lockout, no MFA, an in-process limiter, no CSP or HSTS, no row-level security, a conventionally-immutable audit trail, a temporary admin-login diagnostic, no penetration test and no dependency-audit gate.

8. Design: interface inventory

Screen group	Routes	Purpose	Evidence
Public verification	`/verify`, `/verify-certificate`	Existence and authenticity, one line or a few facts	9 browser checks; 15 API tests
Sign-in	`/login`, `/admin/login`, `/field-officer/login`	Role-specific entry	4 auth tests; browser check 1
Officer workspace	`/field-officer`, `/students`, `/students/register`, `/eligibility`	Register, review the register, score eligibility	23 browser checks; 9 student tests
Award and payment	`/disbursements` (+ certificate panel)	Submit, verify, approve, pay, clear, acknowledge, certify	20 API tests; no browser harness
Administration	`/dashboard`, `/duplicate-review`, `/reports`, `/settings`, `/audit`, `/users`	Adjudicate, monitor, tune, administer	33 API tests; 9 browser checks

Full detail, including the deliberate asymmetry between what an officer and an administrator may see, is in `SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md`. Two interface facts matter for the chapter's honesty: the registry screen renders whatever the endpoint returns because the endpoint has no pagination, and the administrator's award picker has no data because `GET /api/students` is officer-only.

9. Design: database

Aspect	Fact
Tables	13: users, students, duplicate_flags, duplicate_review_events, bursary_awards, bursary_award_verification_events, disbursements, disbursement_reconciliation_events, bursary_certificates, audit_logs, reports, beneficiaries, sbms_schema_migrations
Migrations	14 forward files, dated 20261019 to 20261031, plus one rollback script
Constraints that matter most	approval requires verified; clearance requires evidence; acknowledgement requires officer-assisted method and evidence; one certificate per award; one case per unordered pair; token format `^[a-f0-9]{64}$`
Referential asymmetry	Review data cascades; money, certification and verification data restrict deletion
Indexes	Nine purpose-built indexes; none on student name, school or date of birth, consistent with whole-registry scanning
Legacy	`beneficiaries` table and `scholarship_amount` with unresolved meaning, deliberately not mapped to awards

The full column-level dictionary is `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md` §4. That file is the source for the chapter's Table 4.4 (column design) and must be cited rather than paraphrased.

10. Design: physical deployment

Element	Fact
API	Azure Container Apps, `minReplicas 0`, image `ghcr.io/eggxellent/sbms-api` (1.37 GB)
Frontend	Azure App Service F1 (free), zip deployment of the Vite build
Database	Azure Database for PostgreSQL Flexible Server, B1MS, 32 GiB, stopped between demonstrations
Region	UAE North, forced by a `sys.regionrestriction` policy; Static Web Apps is not creatable on this subscription
Cost	$4.42 per idle month; $4.58 per 8-hour demonstration; about $6.42 in a normal month
Logs	Destination none; diagnostics are the process's own output plus the start-up guard message

Chapter Four's deployment figure must show one container (API plus ML), one web app, one managed database and the public HTTPS ingress, and must state that this is a demonstration topology rather than a production one.

11. What Chapter Four must not claim

Tempting claim	Why it fails
"The architecture scales horizontally"	Rate limits, thresholds and the ML subprocess are per-process; nothing was tested under load
"The design eliminates duplicate payments"	It blocks payment only when a case exists and is open or confirmed; an undetected duplicate still passes
"The ML model was specified as XGBoost"	The proposal specified Random Forest; this is a documented deviation (G-21)
"The system integrates with national registries"	It does not (FR-32)
"Certificates prove payment"	They prove an approval; the PDF says so explicitly

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PART II — EVIDENCE FOR CHAPTER FIVE: IMPLEMENTATION, TESTING AND EVALUATION

12. Implementation environment

Item	Value	Status of the evidence
Operating system	Windows 11 (win32)	Recorded
Node.js	v22.11.0	Recorded
npm	10.9.0	Recorded
Python	3.11.9 with a service virtual environment	Recorded
PostgreSQL	17.11 client and server	Recorded
Docker	CLI 29.9.2, engine **not** running	Recorded; image built by CI instead
Browser automation	Headless Chrome over the DevTools Protocol from Node 22's built-in `WebSocket`	Recorded
Hardware specification	**Not recorded** (G-01); logs show 7.4 GB RAM with about 0.5 GB free	Blocking gap
Test database	Disposable `sbms_test` on loopback, role `sbms_test_runner`, guarded by `testDatabaseGuard.js`	Recorded

The hardware gap is the chapter's weakest environmental statement. It is honest to write that the workstation specification was not captured, that available memory was observed at 7.4 GB with approximately 0.5 GB free during the sweeps, and that one sweep produced 149 or 150 of 151 tests under memory pressure (G-02), rather than to claim a clean environment.

13. Implementation by feature, and where the code is

Feature	Implementation	Tests
Authentication and RBAC	Two sign-in endpoints, JWT, role re-read, user administration	4 + 6 + 1
Registration and validation	Zod schema, five-school constraint, phone and identifier normalisation	9
Duplicate screening	Advisory lock, whole-registry comparison, Python subprocess, `top-1` aggregation, provenance	15 + 13 + 8 + 6 + 1 + 4
Duplicate review	Masked queue, audited reveal, decision with reason, terminal states	9
Eligibility scoring	Five weighted factors, four categories, persisted result	Route and validation only
Award lifecycle	Submission, verification with evidence, approval and rejection with reasons	5
Payments and reconciliation	Issue, clearance with evidence, officer-assisted acknowledgement, discrepancy detection, exceptions	4 + 4 + 4
Student QR identification	32 random bytes, fragment token, existence-only endpoint, dedicated limiter	10 + 4
Award certificates	Snapshot row, reference, token, A4 landscape PDF, revocation, `not_current`	12 + 5
Dashboard, reports, audit, settings	Totals and trend, date-range PDF, audit trail, three cut-offs	6 (authorisation); content by inspection
Operations	Health, readiness, start-up guards, migration ledger	1 + 5 + migration unit tests

14. Test evidence, by level

Level	Command	Result	Date
Backend Jest, whole tree	`npx jest --runInBand tests`	**24 suites / 151 tests pass**	2026-11-01
Backend Jest, earlier generation	`npm test`	21 suites / 128 tests pass	2026-10-31

Level	Command	Result	Date
Backend native, in-process	`node --test` over seven files	**43/43 pass**	2026-10-31
Backend native, out-of-process	`node --test test/api.smoke.test.js`	**3/3 pass**	2026-10-31
Frontend production build	`npm.cmd --prefix frontend run build`	**exit 0**, 2 006 modules, 10.11 s	2026-11-01
ML service	`unittest discover -s ml-service/tests`	**33/33 pass**	2026-10-31
ML research helper	`unittest discover -s experiments/sbms_dataset`	**7/7 pass**	2026-10-31
Browser, registration and review	`node scripts/browser-smoke.mjs`	**23/23 pass**	2026-10-30
Browser, QR page	`node scripts/qr-smoke.mjs`	**9/9 pass**	2026-10-30
Browser, registration fields on the live deployment	`node scripts/registration-fields-smoke.mjs`	**11/11 pass**	2026-11-01
Database restore	`pg_dump` then `pg_restore` into an independent database	Pass, digests compared	2026-10-31
Phone-camera QR scan	manual	**Not run**	—
Browser check of `/verify-certificate`	no harness	**Not run**	—
User acceptance	—	**Not conducted**	—
Load or scale test	—	**Not run**	—
Docker build in this environment	—	**Not run** (engine unavailable)	—

The count that may be quoted as the current automated position is **24 suites / 151 tests, 2026-11-01**, plus the native, browser and Python layers separately. Counts must never be summed across levels, and never quoted without a date.

15. Defects found during testing, and what closed them

Six defects are tabled here because each has a code-level root cause, a fix and a test or browser check at this revision. They are a selection, not a register: the session documents record others, including a date-of-birth feature skew that made two identical dates score `dob_match: 0`, a registration 500 caused by an origin variable read at request time, a deployed SPA that returned 404 for every `/api/verify` call, a review queue that closed a case as soon as it was opened, and a readiness check that passed against a schema missing a table. Those are recorded where they happened, in `docs/SBMS_PROJECT_HANDOVER_2026-10.md` §13.2-§13.4 and `docs/azure-deployment-log-2026-10-31.md` §7-§8, and this report does not restate them.

Defect	Effect before the fix	Root cause	Closed by
Payment clearance returned 500 for every status change	No payment could ever be marked cleared; all reconciliation reporting was unreachable	One SQL parameter reused across `VARCHAR(20)`, `VARCHAR(50)` and a text comparison (PostgreSQL 42P08)	Explicit casts, with a comment recording why they must stay
Stale QR result on a second scan in the same tab	A valid code followed by an invalid one still read "Record found"	Verification ran once on mount; a hash-only change does not remount	`hashchange` listener
StrictMode regression introduced by the first fix	Development runs reported "invalid" for a valid token	The effect re-read an already-stripped URL	Token captured during render, effect keyed on it
`risk_level: critical` unreachable in the UI	A suspicious case could never be styled as such	The API only emitted `review`	API now publishes `priority` and `exact_identifier_matches`
High-priority cut-off was dead configuration	Every model case was `high`; screen showed "high 0.00"	Queue prioritised on the review cut-off and published a null threshold	Priority computed on the high-priority cut-off and published

Defect	Effect before the fix	Root cause	Closed by
Registration form omitted UPI and birth certificate	The critical exact-identifier rule could only fire for seeded rows	Form posted five fields; API accepted eight	Two optional inputs added and verified live with 11 checks

Two of these are the strongest evidence that testing found real faults rather than confirming what was written: payment clearance was completely broken and invisible to the suite until a workflow test exercised it, and the QR defect lived in the interaction between hash routing and React's mount behaviour, where no unit test would have reached it.

16. Machine-learning evaluation

Aspect	Fact	What it supports
Artifact	`xgboost:6ec9f2a3…`, six features, one file	Provenance per case
Holdout	12 000 synthetic pairs; 0 shared groups, 0 shared identities, 0 identical rows	Leakage check
In-sample versus holdout F1	0.9196 versus 0.9063	No memorisation
Grouped cross-validation	0.9067 mean F1, within 0.0004 of the holdout	Stability inside the training file
Learning curve	0.8886 at 10% of groups rising to 0.9063 at 100%	Near-flat, not memorisation
Live cut-off	0.50: precision 0.9204, recall 0.8927, F1 0.9063, miss 10.73%, FPR on hard negatives 7.72%	Synthetic operating point only
Independent FPR baseline	0.35% of 2 000 synthetic unrelated pairs at 0.50	A different question from 7.72%; report both or neither
Per-feature AUC	0.34 to 0.71	No single-feature shortcut
Adjudicated data	**None**	No real-world metric is possible

The reported numbers describe a generator, not Kenyan bursary records. Three limitations carry into Chapter Six: positives are one duplicate pattern (`repeat_application`), the holdout was used for model selection (so the figures are selection-optimistic), and one artifact with one seed means a retrain invalidates them.

17. Evaluation against the objectives

Objective	Status	Evidence	What is missing
(i) Investigate the existing challenges of manual registration and tracking	**Partially achieved**	Chapter Two document analysis; requirement register derived from it	No primary fieldwork or interviews (G-06)
(ii) Design architecture, database schema and ML pipeline	**Achieved with one deviation**	Three-tier architecture, 13 tables, six-feature pipeline, deployed	Random Forest replaced by XGBoost (G-21)
(iii) Implement registration, unique identifiers, QR identification and RBAC	**Achieved**	45 endpoints; QR flow with 10 + 4 tests and 9 browser checks; role matrix with 6 + 11 tests	Phone-camera scan not performed (G-11)
(iv) Validate, test and evaluate effectiveness	**Partially achieved**	151 Jest tests, 43 native, 40 Python, 43 browser checks; synthetic ML evaluation; live deployment checks	No real-user evaluation, no load test, no real-world ML metric

18. Evaluation against non-functional requirements

NFR	Verdict	Basis
NFR-01 performance	Partial	22.67 ms for 50 pairs; no registry-scale measurement
NFR-02 to NFR-06 security and privacy	Verified	Tests plus implementation
NFR-07 to NFR-11 rate limits, bodies, errors, origins	Verified	Tests; in-process limitation recorded

NFR	Verdict	Basis
NFR-12 database integrity	Verified	Constraint tests across workflow and certificate suites
NFR-13 concurrency	Partial	Advisory lock present; no concurrency test
NFR-14 usability	Partial	Browser-checked; no usability study
NFR-15 maintainability	Verified	Shared feature module, ordered migrations, three test runners
NFR-16 zero-cost operation	Partial	Zero-cost configuration is recorded (scale-to-zero, free tiers); no billed month reconciled
NFR-17 human decisions	Verified	Score creates cases only; gates are role-gated
NFR-18 recovery	Partial	Rehearsal done; no schedule or retention
NFR-19 browser compatibility	Partial	Chrome only
NFR-20 degradation	Verified	503 on predictor failure
NFR-21 audit completeness	Partial	Attributed in every tested flow; append-only is convention, not a constraint

In requirement terms this is 14 verified and 7 partial, the same split as §2; the grouped rows above cover NFR-02 to NFR-11 individually.

19. Limits of this evaluation

1. No user acceptance study, so nothing about usability, comprehension or workflow fit can be concluded.
2. No load test, so nothing about concurrency, throughput or latency at scale can be concluded.
3. No real-world duplicate labels, so nothing about model accuracy can be concluded.
4. No phone-camera session, so physical QR identification is unverified even though the code path is tested headlessly.
5. No penetration test or dependency audit, so the security section is a control inventory, not an assurance statement.
6. The deployment is a demonstration instance: one region, no monitoring, no backup schedule, demonstration data.
7. Test counts come from dated runs; none was re-executed for this report.

PART III — EVIDENCE FOR CHAPTER SIX: CONCLUSIONS, RECOMMENDATIONS AND FUTURE WORK

20. Conclusions, objective by objective

Objective	Conclusion the evidence supports	Wording to use
(i) Investigate the challenges	Partially achieved: the problem is characterised from secondary sources, not fieldwork	The investigation establishes the problem from documentary evidence and does not claim primary data
(ii) Design the architecture, schema and pipeline	Achieved, with a stated deviation from the proposal	The artefact is a three-tier system with a 13-table schema and a six-feature pipeline; the classifier is XGBoost rather than the proposed Random Forest, and that change is recorded rather than reconciled
(iii) Implement registration, identifiers, QR identification and RBAC	Achieved	Every capability named in the objective is implemented and covered by executed tests, with the physical scan and the scale behaviour reported as unverified
(iv) Validate, test and evaluate	Partially achieved	The artefact is tested at API, browser and database level and evaluated against synthetic data only; the absence of adjudicated data and of user involvement bounds the conclusion

21. The stated contribution

Contribution	Statement the evidence supports	Evidence
Engineering	A working, deployed bursary system in which an algorithmic signal can never authorise payment, because the gates are enforced in the database as well as the API	`workflow.test.js` #2-#6; schema CHECK constraints
Methodological	The registration-level flag arithmetic: because every registration is compared with the whole registry, a per-pair false-positive rate p produces a nuisance flag with probability 1 − (1 − p)^N, so queue volume grows with registry size independently of cut-off. The `top-1` aggregation policy is the response	`docs/ml-evaluation-baseline-2026-10.md` §3-§4
Evaluation	A reproducible synthetic pipeline with leakage gates, an operating-point sweep, and integrity tests that fail the build if the artifact and its recorded metrics diverge	`docs/ml-training-integrity-2026-10-30.md`; `ml-service/tests/`
Transparency	A documented set of claim limits and stale-document exclusions that a reader can audit	`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`

The aggregation finding is the most defensible contribution because it is arithmetic rather than assertion: it follows from the measured per-pair rate and does not depend on any accuracy claim.

22. Recommendations

#	Recommendation	For whom	Basis
1	Obtain adjudicated duplicate pairs before making any accuracy statement	Programme owner	D1, D5
2	Choose a cut-off only after stating an acceptable miss rate and the weekly review capacity	Programme owner	D2, D3
3	Introduce candidate blocking before the register exceeds a few thousand rows	Project owner	FR-34; the N x p arithmetic
4	Replace the in-process rate limiter with a shared store before running more than one instance	Operator	NFR-07
5	Schedule and retain database backups, and rehearse a restore against the deployment, not only locally	Operator	NFR-18, G-19
6	Perform at least one phone-camera scan of a printed student code and a printed certificate	Author	G-11

#	Recommendation	For whom	Basis
7	Conduct a small structured usability session with officers and record consent	Author	G-06, NFR-14
8	Mark `docs/verification.md` and `docs/uat-test-plan.md` superseded or withdraw them	Author	G-07, G-08
9	Commit the two untracked root reports and resolve the model-artefact ignore rule	Author	G-27, G-28
10	Capture the workstation and device specification, or state explicitly that it was not recorded	Author	G-01

23. Future work

Area	Work	Why it matters	Depends on
Candidate generation	Name-token blocking with a capped candidate list; keep `school_match` as a feature; treat a missing school as a wildcard	Removes the linear and quadratic cost and makes the queue affordable	D4 approval and new tests
Model governance	A registry entry per artifact, with the metrics file bound to the hash	Makes a retrain auditable	The current integrity gate as a starting point
Calibration	Calibrate scores before calling them probabilities, or stop calling them probabilities	Prevents a 0.6 being read as "60% likely a duplicate"	Adjudicated data
Adjudication	In-app review of sampled pairs, stratified by score band, two independent reviewers	Produces the only evidence that can support an accuracy claim	D5
Operations	Scheduled backups, retention, monitoring, alerting and a log workspace	Turns a demonstration into an operated service	Budget decision
Security	Short-lived tokens with refresh, server-side revocation, per-account lockout, MFA for administrators, CSP and HSTS, dependency auditing, a penetration test	Closes the residual risks in the security catalogue §11	Time and review
Data model	Remove the legacy `beneficiaries` module; add the constraint tying a payment's student to its award's student	Removes two parallel registries and one unchecked pairing	Migration and data cleanup
Accessibility	Keyboard and screen-reader audit plus a contrast check	NFR-14 cannot be claimed without it	Design time
Physical identification	Token rotation and re-issue, and a printed-card process	FR-35 and FR-36 are currently gaps	Policy decision on re-issue

24. Boundaries of the conclusions

Boundary	What it excludes
Data boundary	Every metric is synthetic; no claim about Kenyan bursary records is supported
User boundary	No officer, student, guardian or school participated; usability is unmeasured
Scale boundary	Nothing above 50 candidate pairs was timed; no hosted load test exists
Device boundary	Chrome only; no phone camera, no printed output on a physical printer
Operational boundary	Demonstration instance; no monitoring, no backup schedule, no high availability
Security boundary	Control inventory, not an assurance statement; no review or penetration test
Governance boundary	The five owner decisions are unanswered, and no supervisor finding can be closed without its wording

PART IV — ASSURANCE

25. Completeness matrix (requested area to status to location)

Requested area	Status	Where it is covered
Scope, actors and requirement register	Complete	Part I §1-§3; `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §4-§7
Functional requirements with verification status	Complete	Requirements catalogue §6
Non-functional requirements with status	Complete	Requirements catalogue §7
Analysis artefacts and what each must encode	Complete	Part I §4
Architecture, modules and cross-cutting concerns	Complete	Part I §5; `SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md` §1-§10
Access control matrix and security controls	Complete	Part I §7; `SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md` §1-§12
Database schema and full data dictionary	Complete	Part I §9; `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md` §2-§11
API endpoint catalogue with contracts	Complete	`SBMS_API_ENDPOINT_CATALOGUE.md` §2-§7 (45 endpoints)
ML subsystem: features, artifact, metrics, limits	Complete	Part II §16; `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §1-§14
UI screens, workflows and figure mapping	Complete	Part I §8; `SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md` §1-§10
Implementation environment	Complete for software; **gap** for hardware	Part II §12 (G-01)
Test inventory with what each suite proves	Complete	Part II §13-§14; `SBMS_TEST_EVIDENCE_CATALOGUE.md` §2-§11
Defect log and resolution	Complete	Part II §15
Evaluation against objectives and NFRs	Complete	Part II §17-§18
Conclusions, contribution, recommendations, future work	Complete	Part III §20-§24
Claim limits and stale-document exclusions	Complete	Part III §24; `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §3-§5
Figures and screenshots	**Not produced**	`FIGURE_PLACEHOLDER_CHECKLIST.md`; G-03, G-04, G-05
User acceptance evidence	**Not available**	G-06
Physical QR scan evidence	**Not available**	G-11
Load and scale evidence	**Not available**	NFR-01
Production metrics, uptime, monitoring	**Not available**	Part II §19
Supervisor findings SBMS-01…18	**Not recoverable**	`SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md` (repository root) §11

26. Gap summary, ordered by submission risk

Priority	Gap	Effect if unresolved	Owner
Blocking	G-01 workstation and device specification	Chapter Five §5.2.1 cannot be completed	Author

Priority	Gap	Effect if unresolved	Owner
Blocking	G-03/G-04/G-05 figures and screenshots, plus the synthetic-data check	Chapters Four and Five cannot be illustrated	Author
Blocking	G-06 a user-acceptance position, or an explicit statement that none was conducted	NFR-14 stays unverified and Chapter Five §5.5.6 is empty	Author
Important	G-11 one phone-camera scan of a printed code	The identification mechanism rests on a headless-browser check	Author
Important	G-21 the Random Forest to XGBoost deviation	Must be stated in Chapters Four and Six	Author
Important	G-07/G-08 stale `uat-test-plan.md` and `verification.md`	Two documents contradict the implementation	Author
Important	G-02 the 149/150-of-151 instability	Test stability claim needs a caveat	Author
Important	G-25 two test-count generations in one file	A count quoted without a date is misleading	Author
Optional	G-27/G-28 untracked reports and the artefact ignore rule	Reproducibility from a clean clone	Author
Optional	G-19 retention policy, G-17 monitoring	Operational maturity	Operator

27. Reproduction: how a reader can re-establish every claim

```
# 1. Fix the revision under description
git checkout fix/sbms-completion-work
git rev-parse HEAD          # expect d6b5032

# 2. Backend suite (needs backend/.env.test; values are never printed)
cd backend; npm test        # expect 24 suites / 151 tests on 2026-11-01

# 3. Native tests
node --test test/*.node.test.js test/*.node.js test/*.unit.test.js     # expect 43/43
node --test test/api.smoke.test.js                                     # expect 3/3 against a live API

# 4. Frontend build
npm.cmd --prefix frontend run build                                    # expect exit 0

# 5. ML suite and research helper, then the integrity audit and the operating point
cd ..\ml-service
.\.venv\Scripts\python.exe -m unittest discover -s tests -v           # expect 33/33
.\.venv\Scripts\python.exe -m unittest discover -s experiments\sbms_dataset -p test_experiment.py   # expect
7/7
.\.venv\Scripts\python.exe .\check_training_integrity.py
.\.venv\Scripts\python.exe .\evaluate_operating_point.py

# 6. Browser checks (each needs the API, the frontend dev server and a seeded account)
cd ..
node scripts/browser-smoke.mjs                                         # 23 checks
node scripts/qr-smoke.mjs                                              # 9 checks
node scripts/registration-fields-smoke.mjs                             # 11 checks (live deployment)
# node scripts/admin-reveal-smoke.mjs                                   # not re-run: admin password drifted
```

Together steps 5 and 6 reproduce the recorded 40 Python tests and 43 browser checks. The fourth browser harness is listed but commented out deliberately: it cannot pass until the administrator password is reset, and a script that silently skips would misrepresent the evidence.

The Jest guard `backend/tests/testDatabaseGuard.js` refuses to run unless the target database is the disposable loopback `sbms_test`, so step 2 cannot damage the development or deployment database.

28. Document map

File	What it is for	Read it when
This report	The consolidated evidence base and drafting aid	Starting any of Chapters Four to Six
`SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md`	The numbered requirement register with status	Writing requirement tables or conclusion claims
`SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md`	Modules, flows, decisions, weaknesses	Writing the architecture and implementation sections
`SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md`	Every table and column, plus state machines	Writing the database section or an ER diagram
`SBMS_API_ENDPOINT_CATALOGUE.md`	All 45 endpoints with guards and contracts	Writing the interface or API appendix
`SBMS_ML_SUBSYSTEM_DOCUMENTATION.md`	Features, artifact, metrics, limits, decisions	Writing the ML and evaluation sections
`SBMS_TEST_EVIDENCE_CATALOGUE.md`	Every test artefact, date and what it proves	Writing the testing section
`SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md`	Controls, matrix, residual risks	Writing the security section
`SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md`	Resources, pipeline, config, runbook, costs	Writing the deployment section
`SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md`	Screens, workflows, figure mapping	Writing the interface design section or capturing screenshots
`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`	Evidence tiers, forbidden claims, open decisions	Before writing any claim
`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md`	The drafted chapter	Drafting or checking Chapter Four
`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md`	The drafted chapter	Drafting or checking Chapter Five
`CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md`	The drafted chapter	Drafting or checking Chapter Six
`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md`	Objective to requirement to test to conclusion	Defending a conclusion
`DOCUMENTATION_GAP_REGISTER.md`	Every open gap with priority	Before submission
`EVIDENCE_AND_VERIFICATION_REPORT.md`	Code, test and document evidence index	Auditing a specific claim
`FIGURE_PLACEHOLDER_CHECKLIST.md`	What each figure must show	Producing the figures

Entry set added later, same folder. Nine files now form the front door to this evidence base, and each is summarised in `CLAUDE_HANDOFF_GUIDE.md` §2: `ACADEMIC_GUIDE_REQUIREMENTS.md`, `CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN_EVIDENCE.md`, `CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION_EVIDENCE.md`, `CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORK_EVIDENCE.md`, `CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md`, `DIAGRAM_SPECIFICATIONS.md`, `MASTER_EVIDENCE_REGISTER.md` (51 evidence IDs), `ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md` and `CLAUDE_HANDOFF_GUIDE.md`. Read those first; the table above is the deeper tier they point into.

29. Final statement

The evidence supports a clear, bounded conclusion: SBMS is a working, tested, deployed three-tier system whose duplicate-detection signal is deliberately subordinate to human decision, whose payment path is gated by verification and duplicate state in the database as well as the API, and whose machine-learning component is evaluated only against synthetic data. The engineering is real and inspectable. The evaluation is partial, and this documentation set states exactly where it stops: no user acceptance, no load test, no physical QR scan, no real-

world accuracy figure, no production assurance. A report that keeps those boundaries visible is stronger than one that quietly crosses them.